

D847

General Earth Science I

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

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How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A student notes that Earth is closest to the Sun in early January, when the Northern Hemisphere has winter. The student lives near a large lake and loves astronomy. Which explanation of the seasons is correct?

- A. Distance from the Sun, since Earth is closer in summer.
- B. The large lake cools the air in winter.
- C. Earth's tilt changes sunlight angle and day length; the Northern Hemisphere tilts away from the Sun in January.
- D. The Sun gives off less energy in winter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2. [TRANSFER — reversed stem / different variable]

In June, the Northern Hemisphere is tilted toward the Sun. What season is it in Australia, and why?

- A. Winter, because the Southern Hemisphere is tilted away from the Sun.
- B. Summer, because the whole Earth has the same season.
- C. Summer, because Australia is closer to the Sun.
- D. Spring, because it is near the equinox.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3.

A coastal town with a busy fishing harbor records its largest tidal range of the month. The town's lighthouse was built in 1890. What moon phase is most likely?

- A. Only during the season of spring.
- B. New moon or full moon.
- C. Any phase, since tides depend only on the harbor shape.
- D. First or third quarter.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4. [TRANSFER — reversed stem / different variable]

Tides this week have the smallest difference between high and low water. Where are the Sun and Moon relative to Earth?

- A.** Lined up on opposite sides of Earth.
- B.** The Sun has no effect on tides.
- C.** Lined up on the same side of Earth.
- D.** At right angles to each other as seen from Earth.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A city known for its music festival has a record cold week in March. A resident posts that this proves the climate is getting colder. Local weather records go back 120 years. What is wrong with the claim?

- A.** Climate and weather mean the same thing.
- B.** A single cold week is weather; climate trends require decades of averaged data.
- C.** The festival makes the city warmer.
- D.** Nothing; one week is enough to show a trend.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In a mountain park with popular hiking trails, water seeps into cracks in a granite cliff, freezes, expands, and splits the rock. The pieces stay at the base of the cliff. The park receives a million visitors a year. What process is this?

- A.** Chemical weathering.
- B.** Erosion.
- C.** Human erosion from hikers.
- D.** Mechanical weathering (frost wedging).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

Rain washes loose soil from a hillside into a river, which carries it downstream. What process is this?

- A.** Deposition.
- B.** Soil formation.
- C.** Erosion.
- D.** Weathering.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

Along a famous fault near a large coastal city, two plates slide past each other horizontally. The city has many tall glass buildings. What features are expected here?

- A.** Frequent earthquakes with little volcanic activity.
- B.** A chain of subduction volcanoes.
- C.** Volcanoes at every plate boundary.
- D.** A mid-ocean ridge.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9.

A magnitude 6.0 earthquake strikes a region with many old brick buildings. Last year, the same region had a magnitude 4.0 quake. The region is famous for its vineyards. How does the ground-motion amplitude of the 6.0 compare?

- A.** About 1,000 times larger.
- B.** 1.5 times larger.
- C.** 100 times larger.
- D.** 2 times larger.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10. [TRANSFER — reversed stem / different variable]

An earthquake releases about 1,000 times the energy of a magnitude 5.0 quake. What is its approximate magnitude?

- A.** About 1,005
- B.** About 15.0
- C.** About 8.0
- D.** About 7.0

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A volcano on an island known for its coffee farms erupts, sending ash into the air. The ash blocks sunlight, cools the region, and settles on crops, killing some plants. The eruption lasted three days. Which spheres interacted?

- A.** Biosphere only, since coffee farms were affected.
- B.** Geosphere, atmosphere, and biosphere.
- C.** Geosphere only.
- D.** Hydrosphere and atmosphere only.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

12.

A farming valley that hosts an annual harvest fair pumps 5 million cubic meters of groundwater a year. The aquifer recharges 2 million cubic meters a year and currently holds 30 million cubic meters of usable water. At this rate, about how long until the usable water is gone?

- A.** 10 years
- B.** Never, because groundwater is renewable.
- C.** 6 years
- D.** 15 years

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer C Concept 1

A student notes that Earth is closest to the Sun in early January, when the Northern Hemisphere has winter. ~~The student lives near a large lake and loves astronomy.~~ Which explanation of the seasons is correct?

Target word: cause of seasons **Decides:** Closest approach in northern winter rules out distance.

A	Inverted operation	Earth is closest during northern winter.
B	Camouflage	Local water affects climate, not the seasons themselves.
C	Correct	Tilt, not distance, causes seasons.
D	Adjacent category	Solar output is nearly constant through the year.

2. Answer A Concept 1 • transfer

In June, the Northern Hemisphere is tilted toward the Sun. What season is it in Australia, and why?

Target word: opposite hemisphere **Decides:** The Southern Hemisphere tilts away in June.

A	Correct	Opposite tilt gives opposite season.
B	Absolute language	Hemispheres have opposite seasons.
C	Confusion set	Distance does not cause seasons.
D	Adjacent category	June is a solstice, not an equinox.

3. Answer B Concept 2

A coastal town ~~with a busy fishing harbor~~ records its largest tidal range of the month. ~~The town's lighthouse was built in 1890.~~ What moon phase is most likely?

Target word: phase for spring tide **Decides:** Largest range means Sun and Moon are aligned.

A	Confusion set	Spring tides happen twice every month.
B	Correct	Alignment produces spring tides.
C	Absolute language	Local shape matters, but the phase sets spring and neap timing.
D	Inverted operation	Those phases produce neap tides.

4. Answer D Concept 2 • transfer

Tides this week have the smallest difference between high and low water. Where are the Sun and Moon relative to Earth?

Target word: reverse: from tide to geometry **Decides:** Smallest range is a neap tide.

A	Confusion set	Full moon alignment also produces spring tides.
B	Dropped element	The Sun's pull contributes to tides.
C	Inverted operation	That produces spring tides.
D	Correct	Quarter moons produce neap tides.

5. Answer B Concept 3

A city ~~known for its music festival~~ has a record cold week in March. A resident posts that this proves the climate is getting colder. Local weather records go back 120 years. What is wrong with the claim?

Target word: weather vs. climate **Decides:** One week is weather.

A	Confusion set	They differ in time scale.
B	Correct	Climate is the long-term pattern.
C	Camouflage	Irrelevant.
D	Absolute language	Short events do not define climate.

6. Answer D Concept 4

In a mountain park ~~with popular hiking trails~~, water seeps into cracks in a granite cliff, freezes, expands, and splits the rock. The pieces stay at the base of the cliff. ~~The park receives a million visitors a year.~~ What process is this?

Target word: name the process **Decides:** Rock breaks but stays in place.

A	Adjacent category	No change in mineral composition.
B	Confusion set	Erosion moves material away.
C	Camouflage	The visitors are not what split the rock.
D	Correct	Physical breakdown in place.

7. Answer C Concept 4 • transfer

Rain washes loose soil from a hillside into a river, which carries it downstream. What process is this?

Target word: reverse: moving material **Decides:** Material is transported.

A	Adjacent category	Deposition is where material settles.
B	Confusion set	Formation builds soil; this removes it.
C	Correct	Water moves the soil.
D	Inverted operation	Weathering breaks down; it does not move.

8. Answer A Concept 5

Along a famous fault near a large coastal city, two plates slide past each other horizontally. ~~The city has many tall glass buildings.~~ What features are expected here?

Target word: boundary features **Decides:** Horizontal sliding is a transform boundary.

A	Correct	Transform boundaries produce quakes, few volcanoes.
B	Confusion set	Convergent boundaries form those.
C	Absolute language	Transform boundaries rarely have volcanoes.
D	Adjacent category	Divergent boundaries form ridges.

9. Answer C Concept 6

A magnitude 6.0 earthquake strikes a region ~~with many old brick buildings~~. Last year, the same region had a magnitude 4.0 quake. ~~The region is famous for its vineyards~~. How does the ground-motion amplitude of the 6.0 compare?

Target word: amplitude ratio **Decides:** Two steps: 10×10 .

A	Confusion set	That is the approximate energy ratio.
B	Wrong stem number	$6.0 / 4.0$.
C	Correct	$10^2 = 100$.
D	Inverted operation	Divides magnitudes instead of using powers of ten.

10. Answer D Concept 6 • transfer

An earthquake releases about 1,000 times the energy of a magnitude 5.0 quake. What is its approximate magnitude?

Target word: solve for magnitude from energy **Decides:** About $32\times$ per step; $32^2 \approx 1,000$.

A	Wrong stem number	Adds the ratio to the magnitude.
B	Inverted operation	Multiplies instead of using logarithmic steps.
C	Confusion set	Uses the $10\times$ amplitude rule for energy.
D	Correct	Two steps of $\sim 32\times$ give $\sim 1,000\times$.

11. Answer B Concept 7

A volcano on an island ~~known for its coffee farms~~ erupts, sending ash into the air. The ash blocks sunlight, cools the region, and settles on crops, killing some plants. ~~The eruption lasted three days~~. Which spheres interacted?

Target word: trace spheres **Decides:** Geosphere to atmosphere to biosphere.

A	Camouflage	The farms are one effect, not the whole chain.
B	Correct	Eruption, ash in air, crops harmed.
C	Dropped element	Ignores effects on air and life.
D	Adjacent category	Leaves out the source and the crops.

12. Answer A Concept 8

A farming valley ~~that hosts an annual harvest fair~~ pumps 5 million cubic meters of groundwater a year. The aquifer recharges 2 million cubic meters a year and currently holds 30 million cubic meters of usable water. At this rate, about how long until the usable water is gone?

Target word: years to depletion **Decides:** Net loss = $5 - 2 = 3$ million per year.

A	Correct	$30 / 3 = 10$ years.
B	Absolute language	Withdrawal exceeds renewal.
C	Dropped element	$30 / 5$ ignores recharge.
D	Wrong stem number	$30 / 2$ uses recharge alone.

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owned.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	3	
Wrong stem number	A figure copied from the prompt that is not the answer.	3	
Adjacent category	The right idea in the wrong bucket.	6	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	6	
Absolute language	Always, never, guarantee — where judgment applies.	5	
Confusion set	Every option is real; the test is exact role-matching.	9	
Camouflage	Answers a stem fact that was never the question.	4	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Seasons come from axial tilt	Earth's 23.5° tilt changes the angle of sunlight and day length; the hemisphere tilted toward the Sun has summer.	Believing seasons come from Earth's changing distance from the Sun.	Opposite seasons in the two hemispheres, or Earth near perihelion in January.
2	Spring and neap tides	Spring tides (largest range) occur when the Sun, Moon, and Earth align at new and full moon; neap tides (smallest range) occur at first and third quarter.	Linking spring tides to the season of spring.	A moon phase and a question about tidal range.
3	Weather vs. climate	Weather is short-term atmospheric conditions; climate is the long-term average pattern, typically over 30 years.	Using one cold day or season to judge climate.	A single event compared with a long-term trend.
4	Weathering vs. erosion	Weathering breaks rock down in place (mechanically or chemically); erosion moves the pieces by water, wind, ice, or gravity.	Using weathering and erosion as synonyms.	Whether material stays in place or is carried away.
5	Plate boundaries and their features	Convergent boundaries form subduction volcanoes, trenches, and mountains; divergent boundaries form rifts and ridges; transform boundaries produce earthquakes but little volcanism.	Expecting volcanoes at every plate boundary.	Plates sliding past, colliding, or separating.
6	Earthquake magnitude is logarithmic	Each whole-number step in magnitude means 10 times the ground-motion amplitude and about 32 times the energy released.	Treating magnitude 6 as twice magnitude 3.	Two magnitudes compared.
7	Earth system interactions	Events move matter and energy among the geosphere, hydrosphere, atmosphere, and biosphere; tracing an event means naming each sphere affected.	Naming only the sphere where the event started.	A single event with effects in air, water, land, or life.
8	Sustainable use of groundwater	A resource is used sustainably only if withdrawal does not exceed renewal; an aquifer pumped faster than recharge is being depleted.	Treating any naturally renewing resource as unlimited.	Withdrawal and recharge rates given together.

Course scope (public description): This comprehensive survey course provides a foundational understanding of Earth's position in the universe and its dynamic systems. Students will explore Earth's placement within the solar system, including the effects of its orbit, origin, and celestial interactions on tides and seasons. The course delves into Earth's subsystems—geosphere, hydrosphere, biosphere, and atmosphere—focusing on their interactions and impacts on weather, climate, and geological processes like weathering, erosion, and soil formation. Additionally,

the course examines natural phenomena such as earthquakes and volcanoes, and the significant influence of human activities on Earth's systems, addressing topics such as resource management, land use, pollution, and sustainability. Through interactive simulations, multimedia resources, and real-life applications, students will engage in interdisciplinary thinking and develop a deep appreciation for the scientific methods, theories, and laws that underpin Earth science.